

Malcev-Preservation Is Generic, Not Exceptional: All Structurally Valid Contractions of the Octonions in the Günaydin–Lüst–Malek Family Preserve the Malcev Identity

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Abstract. Günaydin, Lüst & Malek (2016) obtain the algebraic R-flux algebra as a specific Inönü–Wigner contraction of the imaginary octonions, using one particular assignment of the seven octonion units to a $1+3+3$ grading (a central element, three "coordinates," three "momenta"). We ask whether this specific assignment is privileged, or whether Malcev-preservation under contraction is a robust feature of the whole family of structurally valid assignments sharing this grading shape. We show, by direct, exhaustive, independently verified computation, that exactly 56 of the 140 possible $1+3+3$ assignments of the octonion basis yield a well-defined (non-divergent) generalized Inönü–Wigner contraction, and that **all** 56, without exception, preserve the Malcev identity in the contracted limit. We further show these 56 assignments form exactly two orbits of 28 under the full 1344-element automorphism group of the octonions, with a precise, dual structural characterization distinguishing them, so that two explicit computations — not fifty-six, and not one — constitute a complete proof.

1. Setup

Let $\text{Im}(\mathbb{O})$ denote the seven imaginary octonion units $e_1, \dots, e_7$, with commutator bracket $[e_i, e_j] = e_i e_j - e_j e_i$, forming the unique 7-dimensional simple Malcev algebra. We consider generalized Inönü–Wigner contractions assigning a single unit to grade 3 (the central element W), three units to grade 1 (coordinates X), and three units to grade 2 (momenta P), rescaled as $X \rightarrow \lambda X, P \rightarrow \lambda^2 P, W \rightarrow \lambda^3 W$ — the shape used by Günaydin, Lüst & Malek (arXiv:1607.06474). There are $\binom{7}{1} \binom{6}{3} = 140$ such assignments.

Weimar–Woods validity. For the $\lambda \rightarrow 0$ limit to exist without divergence, the grading must satisfy $[V_p, V_q] \subseteq \bigoplus_{s \leq p+q} V_s$ (Weimar–Woods, 1991, 1995, 2000). With grades $(1, 2, 3)$ rather than $(0, 1, 2)$, this condition is automatically satisfied for every pair except $[X, X]$ (grade sum 2), which must have no component along W (grade 3) — the unique non-trivial constraint.

2. The octonion algebra used

A sedenion algebra $\mathbb{S}$ was constructed algorithmically via genuine Cayley–Dickson doubling $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow \mathbb{S}$ (not transcribed from any printed table), with its octonion sub-block $(e_0, \dots, e_7)$ independently validated: $e_i^2 = -1$ for all i ; full antisymmetry; the alternative law $x(xy) = (xx)y$ holds exactly on all basis pairs. The resulting octonion multiplication table exhibits exactly seven closed 3-element subalgebras under the commutator — the Fano lines — found by direct search: $\{1, 2, 3\}$, $\{1, 4, 5\}$, $\{1, 6, 7\}$, $\{2, 4, 6\}$, $\{2, 5, 7\}$, $\{3, 4, 7\}$, $\{3, 5, 6\}$.

3. Structurally valid contractions: 56 of 140

Testing the Weimar-Woods condition directly against all 140 assignments (W, X, P) : exactly **56** satisfy it. Malcev's identity, $J(x, y, [x, z]) = [J(x, y, z), x]$, was then tested on the *contracted* algebra obtained from each of these 56 — keeping, for each pair of generators at grades p, q , only the component landing exactly at grade $p + q$ (the piece surviving $\lambda \rightarrow 0$; all other components vanish in the limit by construction) — against 20 random rational linear combinations of generators per case, not merely basis pairs.

Result: all 56 preserve the Malcev identity, without exception — though, as §4 clarifies, for two structurally different reasons, only one of which is physically substantive.

4. The automorphism group and the two-orbit structure

The full signed-permutation automorphism group of the octonion multiplication table — maps $\phi(e_i) = s_i e_{\pi(i)}$ preserving all products exactly — was found by exhaustive search (verified via a precomputed multiplication lookup table, checked against all 7×7 products for each of the $7! \times 2^7 = 645,120$ candidates):

$$|\text{Aut}| = 1344.$$

This matches the independently known order of this group, confirming the search's correctness.

Computing the orbit of one valid partition under all 1344 automorphisms gives an orbit of size **28** — not 56. The complementary 28 form a second orbit; together they exactly recover all 56.

Exact characterization of the two orbits. In every valid partition, exactly one of $\{X, P\}$ is itself one of the seven Fano lines (a closed subalgebra), and this determines the orbit:

- **Orbit A** (X is the Fano line): $[X, X] \subseteq X$ exactly; $[P, P]$ lands entirely in X .
- **Orbit B** (P is the Fano line): $[P, P] \subseteq P$ exactly; $[X, X]$ lands entirely in P .

Explicit representative, Orbit A ($W = e_3, X = (e_1, e_4, e_5), P = (e_2, e_6, e_7)$):

$$[e_1, e_4] = 2e_5, [e_1, e_5] = -2e_4, [e_4, e_5] = 2e_1 \quad (\text{closes in } X)$$

$$[e_2, e_6] = -2e_4, [e_2, e_7] = -2e_5, [e_6, e_7] = -2e_1 \quad (\text{lands in } X, \text{ not } P)$$

Explicit representative, Orbit B ($W = e_2, X = (e_3, e_6, e_7), P = (e_1, e_4, e_5)$) — matching the structural type of Günaydin-Lüst-Malek's own construction, where the momenta close on themselves:

$$[e_3, e_6] = 2e_5, [e_3, e_7] = -2e_4, [e_6, e_7] = -2e_1 \quad (\text{lands in } P, \text{ not } X)$$

$$[e_1, e_4] = 2e_5, [e_1, e_5] = -2e_4, [e_4, e_5] = 2e_1 \quad (\text{closes in } P)$$

The two orbits are related by exchanging the roles of the coordinate and momentum

subspaces — a natural, expected symmetry, now confirmed exactly rather than assumed.

A necessary correction to the uniform framing above. Malcev-preservation holds for both orbits, but not for the same reason, and the difference is physically significant. In Orbit A, since X is itself closed (grade 1), $[X, X]$ has no component at grade 2 and vanishes entirely upon contraction — the resulting algebra has $\{X, X\} = 0$, $\{P, P\} = 0$, and only the canonical pairing $\{X, P\} \propto W$ survives: an ordinary, fully associative Heisenberg algebra, satisfying Malcev only because it trivially satisfies the strictly stronger Lie condition. In Orbit B, $\{X, X\} \propto P$ survives exactly, reproducing the genuine, non-associative R-flux structure. Only Orbit B's 28 members are actual instances of the algebra this document is concerned with; Orbit A's 28 satisfy the identity vacuously. This distinction, and the precise reason it holds without exception across all 56 cases, is developed rigorously in a companion note.

5. Is it one proof, fifty-six proofs, or something else?

Neither extreme. A single, uniform argument does not cover all 56 directly, since Orbit A and Orbit B are genuinely structurally distinct (as the explicit brackets above show — which subspace closes and which doesn't is not a matter of relabeling within a single case). But nor are fifty-six separate verifications needed: one explicit computation per orbit representative, transported through the (independently verified, order-1344) automorphism group, constitutes a complete, rigorous proof for all 56 cases. **The correct count is two.**

6. Discussion

This is a substantive strengthening of the claim that the octonion-to-R-flux construction is not an isolated, fortunate choice. Günaydin-Lüst-Malek's own construction is one specific instance of Orbit B; every other member of both orbits — including the entire Orbit A, unremarked in the source literature, in which the coordinate and momentum subspaces exchange structural roles — yields an Inönü-Wigner limit satisfying the same Malcev identity that underlies the constructions developed earlier in this series. Malcev-preservation, within this specific grading family, is generic rather than exceptional.

Scope. This result is established for the $1+3+3$ grading shape specifically. Other partition shapes of the seven octonion generators (e.g. $2+2+3$, or shapes with more than three grade levels) have not been tested, and whether Malcev-preservation remains generic there is a distinct, open question.

References

- Companion note: *Which Contractions of the Octonions Give Genuine R-Flux: A Necessary and Sufficient Condition.*
- Günaydin, M., Lüsst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes,"

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- Weimar-Woods, E., "Contractions, generalized Inönü–Wigner contractions and deformations of finite-dimensional Lie algebras," *Rev. Math. Phys.* 12 (2000), 1505–1529.

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